

Design of Deployment-Controllable Hinge Mechanism



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Abstract

Deployable hinge mechanisms are critical for lightweight folding structures, yet uncontrolled deployment can induce excessive kinetic shocks. This study presents the design of a controllable deployable hinge mechanism incorporating an external modular sleeve integrated with a friction-based damping module to achieve controlled deceleration. The 3D geometry and mechanical assembly were developed and analyzed using SolidWorks. Preliminary estimations demonstrate that the proposed design achieves a 6.7% deceleration rate per single-side unit, effectively mitigating deployment impact while maintaining structural simplicity. This mechanism offers a cost-effective and retrofittable solution for precision deployment systems.

Work Flow

Literature Review

Design Requirements

Methodology Selection

CAD Modeling & Design

Literature Review

Literature review identified various approaches for controlling deployment motion, including thermal actuation, escapement mechanisms, flexure-based mechanisms, and friction-based damping. It also provided methods for estimating the **required deployment torque** for spacecraft deployable structures.

Design Requirements

(Based on ECSS-E-ST-33-01C)

Minimum Actuation Torque:

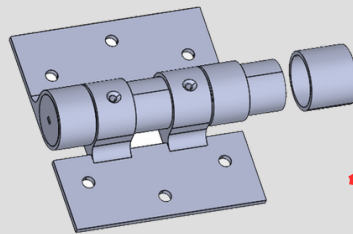
Given

$$m_{panel} = 0.4kg, \quad L = 0.366m, \quad F_R = 0.1N \cdot m$$

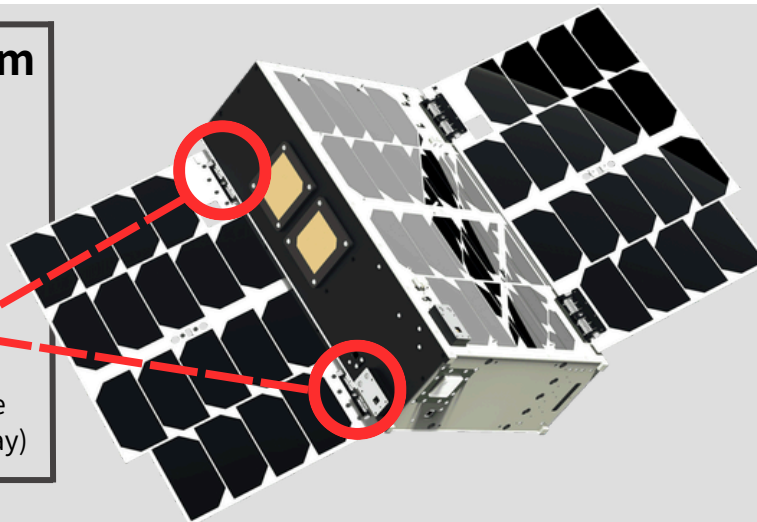
$$\rightarrow T_{min,panel} \approx 0.60N \cdot m$$

(FR dominates > 99% of Tmin — motivates **friction** as the primary tunable parameter)

- Deployment mechanism $\geq 0.60 N \cdot m$
- Controlled Deployment
- Rigid Latch-Lock
- Standard 6U Fit



(Application context: deployment hinge location on a 6U CubeSat solar panel array)



Methodology Selection

Control Type	Approach	Pros	Cons
Active	Motor	Precise, programmable	Power consumption, additional volume & mass
	SMA	Compact, lightweight	Thermal response, limited controllability
Passive	Magnetic Damping	Contactless, low wear	Additional components, limited tunability
	Friction Damping	Simple, compact, tunable	Wear, parameter-dependent

Selection Logic

CubeSat Envelope Constraint
Limited volume & mass

↓
Design Priority
No additional actuator
Passive & compact

↓
Evaluation Result
Friction damping best satisfies the requirements with minimal additional volume.

↓
Selected Method
Passive Friction Damping



CAD Modeling

